Rajagopal Somaskandan

+91 7200288242 | somaskandan931@gmail.com | linkedin.com/in/somaskandan-rajagopal |
rajagopal-somaskandan.netlify.app

Summary

Entry-level Machine Learning Engineer with hands-on experience building end-to-end AI systems using FastAPI, PyTorch, and modern ML frameworks. Built production-oriented projects in fraud detection, recommendation systems, computer vision, and privacy-preserving crowd analytics. Strong interest in backend ML systems, explainable AI, and scalable deployment.

Education

Sathyabama Institute of Science and Technology — Expected 2027
M.Sc. Computer Science

Sathyabama Institute of Science and Technology — 2025
B.Sc. Computer Science (AI Specialization)

Technical Skills

Programming Languages: Python, Java, SQL
Machine Learning: Supervised Learning, Unsupervised Learning, Feature Engineering, Model Evaluation, Explainable AI
Deep Learning: PyTorch, Neural Networks, CNNs, RNNs, LSTMs, GRUs
NLP & Retrieval: Transformers, Embeddings, RAG, BERT, Sentence-Transformers, FAISS, ChromaDB
Backend & APIs: FastAPI, REST APIs, Streamlit
Computer Vision & 3D: OpenCV, Optical Flow, COLMAP, 3D Gaussian Splatting, FFmpeg
Databases: PostgreSQL, MongoDB
Tools & Platforms: Git, GitHub, Docker, Linux, React

Experience

Production AI Developer (Intern) — Mar 2026 – Jun 2026
InnoXR Labs Pvt. Ltd. — *Chennai, India*

- Built and deployed ML-powered FastAPI microservices for real-time inference in AR and immersive applications.
- Designed API-first ML systems integrating model inference with scalable backend endpoints.
- Delivered end-to-end features from model development to deployment in a startup environment.

Data Analyst Intern — Jul 2024 – Oct 2024
India Advocacy — *Chennai, India*

- Built Java and Selenium pipelines to scrape and process **1M+ records**, transforming raw data into **15–20k structured datasets** through multi-stage validation.

Security Trooper (National Service) — Jan 2020 – Jan 2022
Singapore Armed Forces — *Singapore*

- Executed security operations in regulated environments, ensuring strict protocol compliance and operational readiness.
- Supported COVID-19 response initiatives through structured data collection and reporting under time-sensitive conditions.

Sales Associate (Part-Time) — Feb 2022 – Aug 2022
Bata — *Singapore*

- Handled high-volume customer interactions in a fast-paced retail environment, ensuring efficient service delivery.
- Managed billing, inventory coordination, and issue resolution to support smooth store operations.

Projects

FinGuard Pro: Real-Time Fraud Detection & Compliance System — IEEE ICAAIC 2025

- Built a fraud detection system using **XGBoost**, achieving **94% recall** and **ROC-AUC 0.995** on highly imbalanced financial data.
- Integrated **SHAP-based explainability** to generate audit-ready insights for AML and compliance workflows.
- Deployed backend services with **FastAPI** and a **Streamlit dashboard** for real-time monitoring and alerts.
- Automated report generation in PDF and ZIP formats to reduce manual investigation effort.

CareerGenie: AI Career Platform Hackathon Winner, Origin 2026

- Developed an AI-powered career platform using **RAG** with **MiniLM embeddings** and **ChromaDB** for semantic job matching.
- Implemented personalized recommendations using **learning-to-rank (BPR)** based on user feedback signals.
- Built a multi-agent pipeline with **LLaMA** to improve response quality and reduce hallucinations.
- Designed a full-stack system with a **FastAPI backend** and **React frontend**.

MonoSplat: Single-Video 3D Reconstruction Pipeline Internship Project, 2026

- Engineered a 3D reconstruction pipeline using **COLMAP** and **Gaussian Splatting** in **PyTorch**.
- Designed a hybrid **CPU-GPU pipeline**, optimizing preprocessing with **FFmpeg** for cloud execution.
- Built an async **FastAPI** backend with a **Three.js** viewer for interactive 3D visualization.
- Focused on performance and usability, enabling smooth rendering and faster end-to-end processing.

S.A.F.E: Privacy-Preserving Crowd Anomaly Detection Design Thinking Project, 2026

- Developed a privacy-first crowd anomaly detection system using **Farneback optical flow** and zone-based aggregation to analyze motion without tracking individuals.
- Evaluated unsupervised models including **Isolation Forest**, **One-Class SVM**, **Z-Score**, **Moving Average Detector (MAD)**, and **LSTM Autoencoder**; **MAD** achieved strong cross-domain performance with **ROC-AUC 0.8391** on the Mall dataset.
- Reduced false positives by **40–60%** using camera shake detection based on global motion thresholds.
- Built a **FastAPI** backend and **Streamlit** dashboard that provide explainable, real-time alerts for safety operators.

SourceUp: Explainable Supplier Recommendation System Final Year Capstone, 2026

- Implemented semantic supplier search using **Sentence-BERT** and **FAISS** for natural language queries.
- Designed a ranking model using **Scikit-learn** with cost, delivery time, and reliability constraints.
- Added an explainability layer to provide feature-level insights for supplier selection and risk assessment.

Publications & Conferences

FinGuard Pro: Explainable AI for Financial Fraud Detection | *ICAAIC 2025* ISBN: 979-8-3315-6587-9
Real-Time Explainable Fake News Detection using BERT-LSTM | *ICASET 2025*

Achievements & Hackathons

Industry Innovation Hackathon | *Sathyabama Institute of Science and Technology – National Level* Mar 2026

- Won **Best Industry Innovation Award** at Industry Innovation '26.

Origin 24-Hour Hackathon | *SIMATS Engineering, Chennai* Apr 2026

- Placed **4th** at Origin 24-Hour Hackathon.